

MDS-UPDRS Finger Tapping Analysis

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [125, 142, 112, 138, 119, 120, 125, 110, 126, 106]

Tap amplitudes (pixels): [170.94151046483705, 166.42415690037308, 161.31955864060626, 159.81239000778382, 148.66068747318505, 160.86329600005092, 146.60491124106312, 135.14806694880988, 140.0142849854971, 146.64924138910504, 144.42298985964803]

Gemini 3 Pro

[HTML Chart Displayed via UI]

Based on the extracted features provided, here is the analysis and scoring according to the MDS-UPDRS Item 3.4 criteria:

Analysis

1. Rhythm (Tap Periods):

The tap periods (in frames) are: [125, 142, 112, 138, 119, 120, 125, 110, 126, 106].

- The rhythm is irregular. There are notable fluctuations in speed, specifically the jumps to **142** and **138** frames (compared to the faster taps of ~110-112).
- These slower taps can be interpreted as interruptions or hesitations. There are approximately 2 to 3 distinct hesitations (at taps 2, 4, and possibly 9).
- This pattern aligns with **Score 1** ("one or two interruptions") or borderline **Score 2** ("3 to 5 interruptions"). There is no consistent "slowing" trend (the end is actually faster than the beginning).

2. Amplitude (Tap Amplitudes):

The tap amplitudes (in pixels) are: [170.9, 166.4, 161.3, 159.8, 148.7, 160.9, 146.6, 135.1, 140.0, 146.6, 144.4].

- There is a clear, continuous decrement in amplitude at the beginning of the sequence:
 - Tap 1: 170.9
 - Tap 2: 166.4 (Decrease)
 - Tap 3: 161.3 (Decrease)

- Tap 4: 159.8 (Decrease)
- Tap 5: 148.7 (Decrease)
- The amplitude drops consistently from the very first tap through the fifth tap.
- While there is some recovery in amplitude at Tap 6, the initial sequence demonstrates a decrement that begins immediately.
- This specific pattern matches the criterion for **Score 3**: *"The amplitude decrements starting after the 1st tap"*.

Scoring Rationale

The MDS-UPDRS scoring instructions state that a score is assigned if **any** of the listed criteria are met.

- **Score 2 Criteria:** "Amplitude decrements midway" or "3 to 5 interruptions".
- **Score 3 Criteria:** "The amplitude decrements starting after the 1st tap".

Since the amplitude begins to decline immediately after the first tap (170 -> 166 -> 161...), the performance meets the specific criterion for Score 3.